Cost Benefit Memorandum

Model Selection for the Emergency Department Triage Tool, Phase 3

To: Dr De Freitas · the ED Board · Martina Griffith (Clinical IT Lead, Mercer IT Governance) **From:** Shakada Blake, CariSurg MedTech Pathways **Date:** 21 July 2026 **Subject:** Whether to carry a more complex model into Phase 3, and if so, which one.

Verdict

Adopt the **Random Forest** as the Phase 3 triage model. It identifies twice as many critically ill patients as the Week 6 baseline, and under-triage is the costliest error an emergency department can make. The Board should approve this knowing it costs us materially in overall accuracy, a trade we make deliberately, and one this memo sets out in full.

> **AT A GLANCE Decision sought:** approval to adopt the Random Forest for Phase 3. **What we gain:** eight of sixteen critical patients identified, against four under the current baseline. **What it costs:** overall accuracy falls from 0.667 to 0.491, and the model raises roughly fifty-four false ESI-1 alarms. **What it does not settle:** the result rests on sixteen patients and has not been tested on Caribbean data.

Dataset and methods

Three principles govern the comparison: the models see only what a triage nurse sees, every model is tested on identical patients, and the scoring reflects clinical risk rather than headline accuracy.

- **Data.** A de-identified emergency-department dataset of 55,121 attendances. Each record is one arrival; the label is the Emergency Severity Index (ESI) , a five-level triage scale where level 1 is most urgent and level 5 least urgent.
- **Inputs.** 208 numerical features, all available at the moment of triage: vital signs and recorded chief complaints. Demographic, administrative and index columns are excluded, as are fields recording what happened after triage (such as admission or discharge), which would let the model "see the future", an error known as data leakage.
- **Like-for-like testing.** Every model is trained on the same 44,096 patients and tested on the same 11,025, stratified by ESI level, using the fixed random seed (42) carried over unchanged from Week 6.
- **How we score.** Precision, recall and F1 are reported as macro averages, so each ESI level counts equally rather than being drowned out by the common mid-range levels. Our headline metric is recall for ESI-1: the share of genuinely critical patients the model correctly flags.

- **A caution on scale.** ESI-1 patients make up roughly 0.1% of arrivals, just **16 patients in the test set**. Every ESI-1 figure below rests on that small group and should be read as indicative rather than settled.

The benchmark

Axis	**Dummy**	**Logistic Regression (baseline)**	**Decision Tree (baseline)**	**Random Forest (recommended)**	**Gradient Boosting (compared)**
Accuracy	0.375	0.667	0.556	0.491	**0.668**
Precision (macro)	0.204	**0.582**	0.265	0.403	0.496
Recall (macro)	0.204	0.463	0.245	**0.553**	0.435
F1 (macro)	0.204	**0.492**	0.216	0.390	0.442
Recall - ESI-1	0.000	0.250	0.000	**0.500**	0.125
Training time	0.0 s	8.8 s	0.3 s	15.2 s	20.7 s
Inference (per prediction)	0.0004 ms	0.0041 ms	0.0006 ms	0.0570 ms	0.0708 ms
Interpretability	N/A	High	High	Medium	Low

Of the 16 critical arrivals in the test set: the Random Forest caught 8, logistic regression 4, gradient boosting 2, and the decision tree none.

What the numbers show

The headline finding is uncomfortable for anyone who reads accuracy alone. Gradient boosting posts the best overall score in the table 0.668 and yet it identifies only **two** of the sixteen critical patients in the test set. A model can look strong and still be unsafe at triage, because ESI-1 arrivals are so rare that missing all of them barely moves the accuracy figure.

The Random Forest reverses that pattern. Balanced class weighting pushes it to attend to rare, high-acuity patients, and it identifies eight of sixteen double the baseline. The cost is visible in the same table: overall accuracy falls to 0.491, and macro F1 drops below the baseline. The forest buys recall by spending precision.

How to read the benchmark. Accuracy is the share of all patients given the correct level. Precision asks, of the patients the model labelled ESI-*n*, how many truly were; recall asks, of the patients who truly were ESI-*n*, how many the model caught. F1 balances the two. All three are macro-averaged across the five levels. Training time is a one-off cost; inference time is paid on every patient, at the bedside.

Inside the recommended model

A single accuracy figure hides how a triage model behaves level by level. The Random Forest, opened up across all five acuity levels:

ESI level	Precision	Recall	F1	Patients (test)
1 - critical	**0.13**	**0.50**	**0.21**	**16**
2 - emergent	0.63	0.64	0.63	3,585
3 - urgent	0.80	0.30	0.44	5,402
4 - less urgent	0.31	0.77	0.45	1,779
5 - non-urgent	0.15	0.56	0.23	243
Macro average	0.40	0.55	0.39	11,025

Two patterns matter for the Board. First, the model is deliberately biased towards **sensitivity over precision** at the urgent end: it recovers half of ESI-1 patients and 77% of ESI-4, but its ESI-1 precision is only 0.13, so most ESI-1 alerts are false. That is the balanced-weighting trade working as designed in triage, a false alarm is cheaper than a missed emergency. Second, its weakest recall is at **ESI-3** (0.30): mid-acuity patients are frequently pushed up or down a level, which is clinically safer than the reverse but adds review load.

What drives a prediction

The same run surfaces which inputs the model relies on the evidence behind its "Medium" interpretability rating, and the raw material for the bedside Explainability Panel.

#	Global driver (feature importance)	Weight	Worked example: why one patient was rated ESI-2 (SHAP)
1	Stroke-alert flag	0.113	Shortness of breath (+0.082)
2	Systolic blood pressure	0.087	Oxygen-delivery device (+0.074)
3	Oxygen-delivery device	0.076	Oxygen saturation (+0.034)
4	Diastolic blood pressure	0.072	Systolic blood pressure (+0.020)
5	Abdominal-pain flag	0.065	Heart rate (+0.013)

6	Respiratory rate	0.055	Respiratory rate (+0.013)
7	Heart rate	0.050	Temperature (-0.012, pushes away)

The global drivers are clinically face-valid: a stroke-alert flag, blood pressure, oxygen-delivery device and respiratory rate lead the ranking the same signals a triage nurse reads. For an individual, SHAP decomposes the decision in the same vocabulary, and the explanation took **0.2 seconds** to produce, comfortably inside the one-minute test.

(Figures 1 and 2 in the PDF version chart the ESI-1 comparison and the accuracy-versus-recall trade-off.)

Three arguments for the Random Forest

- **It catches twice as many critical patients as the baseline.** ESI-1 recall rises from 0.250 to 0.500 eight of sixteen critical arrivals identified, against four for logistic regression and two for gradient boosting. It also posts the best macro recall of any model (0.553), meaning it performs most evenly across all five acuity levels rather than favouring the common ones.
- **It remains fast enough for any realistic deployment.** Training completes in about fifteen seconds and each prediction takes 0.057 milliseconds roughly seventeen thousand patients per second on ordinary hardware. There is no hidden computer bill, and no specialist infrastructure is required.
- **It is explainable at the bedside.** The model's strongest drivers are clinically recognisable: stroke alert, systolic and diastolic blood pressure, oxygen delivery device, abdominal pain and respiratory rate. For an individual patient, a SHAP explanation was produced in **0.2 seconds** well inside the one-minute test. In a worked example, the model's ESI-2 rating was traced to shortness of breath, oxygen delivery device and oxygen saturation, which is precisely the "Explainability Panel" capability specified in our Week 4 proposal.

Three arguments against it

- **Overall accuracy falls sharply.** At 0.491, the Random Forest is markedly less accurate than both the logistic-regression baseline (0.667) and gradient boosting (0.668). Roughly half of all patients receive the wrong acuity level. Its macro F1 (0.390) also sits below the baseline (0.492): the balanced class weighting buys recall by sacrificing precision.
- **It generates a heavy false-alarm load at ESI-1.** Precision for the critical class is only 0.13. To catch those eight genuinely critical patients, the model flags roughly sixty as ESI-1 around fifty-four false alarms. In a department already under strain, that alert burden must be planned for, or staff will begin to ignore it.

- **The headline gain rests on sixteen patients.** The difference between the Random Forest and the baseline is four correctly identified patients. That margin is too small to treat as a stable estimate of real-world performance.

Risks and unknowns

- **Sample size at the critical class.** With only 16 ESI-1 patients in the test set, our headline metric is statistically fragile. A larger sample, or repeated cross-validation, is needed before the result can be relied upon.
- **Transfer risk.** The data originates from a US academic hospital. Whether these triage patterns hold in a Caribbean emergency department is untested and must be validated on local Mercer data before any clinical use.
- **Residual under-triage.** Even at its best, the model misses half of all critical arrivals. It is decision support, not a replacement for clinical judgement.
- **Fairness, drift and calibration.** Performance across demographic groups has not been audited, accuracy may drift as the patient mix changes, and we have not yet agreed with clinicians how a borderline prediction should be acted upon.

Recommendation

Adopt the Random Forest for Phase 3, subject to four conditions:

- Re-estimate ESI-1 performance on a larger sample, or through repeated cross-validation, before any clinical pilot.
- Agree an alert-handling protocol with ED staff that accounts explicitly for the false-alarm rate.
- Validate the model on local Mercer data prior to deployment.
- Monitor for fairness and drift from the first day of operation.

What this choice does not solve. It does not close the under-triage gap; half of critical patients are still missed. It does not deliver the best overall accuracy; gradient boosting does, and we have deliberately declined it because a model that identifies two critical patients in sixteen is unsafe at triage regardless of its headline score. It does not prove the model transfers to a Caribbean setting. It does not, on its own, guarantee resilience when biometric fields are missing the founding concern of this programme which remains to be stress-tested. And it does not remove the need for clinician oversight. The Random Forest is the most defensible balance of clinical safety, cost and explainability given what we know today: a decision-support aid to be piloted and monitored, not a finished clinical instrument.